

ABHIJITH RAGAV

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EDUCATION

Georgia Institute of Technology

Aug 2021 – May 2023

Master of Science in Computer Science

Courses: Machine Learning, Artificial Intelligence, Big Data Systems & Analytics, Deep Learning, Ubiquitous Computing

SRM Institute of Science and Technology

Jul 2016 – Jun 2020

Bachelor of Technology in Information Technology

Relevant Courses: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Probability & Statistics, Object Oriented Analysis and Design, Data Mining

EXPERIENCE

Amazon, Inc

Seattle, WA

Applied Scientist II — Buyer Risk Prevention

Jan 2024 – Present

→ Build end-to-end models that protect checkout and prevent fraudulent transactions and bad actors on Amazon.com; blend temporal graph learning over customer–device–payment–address interactions (dynamic embeddings to surface collusive clusters and synthetic identities) with transformer models for tabular/anomaly signals and gradient-boosted trees for calibrated decisioning, plus Bedrock-powered LLM workflows for investigator assist.

→ **Led the development and global deployment of a unified account-takeover (ATO) prevention capability** that generalizes across diverse global marketplaces — built a large-scale entity-relationship graph with multi-hop relational reasoning and Temporal GNN embeddings, combined with region-aware tabular features in a hybrid production architecture; owned end-to-end (modeling, infrastructure, production safeguards), delivering **over \$10M in annual savings** while maintaining minuscule false-positive rates.

→ Drove an **~70% reduction in the bad-debt rate** (tens of millions of dollars in business impact) and a double-digit reduction in orders flagged for manual investigation, improving investigator efficiency and handle time.

→ Built an autonomous **adversarial red-teaming agent** (reinforcement learning with **GRPO**) that probes production fraud-decisioning systems to discover evasion paths and weaknesses before they can be exploited, feeding the surfaced gaps back into more robust, defensible defenses.

→ Built **generative-AI tooling** that became high-leverage across the org: a **code-automation agent** (QLoRA fine-tuning + retrieval-augmented generation over curated internal specs, with validation guardrails) that turns natural-language intent into framework-compliant model-input logic, **improving authoring productivity by over 50%**; and an audit / regulatory-reporting automation that reduced roughly a week of manual compilation and validation to hours while preserving accuracy and auditability.

→ Built an **agentic temporal knowledge graph** in **NebulaGraph** that profiles individual customer journeys from raw **browsing behavior** — turning millions of low-level events (searches, product views, reviews, purchases, etc) into a connected story of who a customer is, what they care about, and how their intent evolves over time. Layered **agentic LLM-driven persona** with **agentic RAG** on top to distill each customer into canonical archetypes and fine-grained interests, surfacing behavioral signals (consistency, drift, anomalies) that power risk, trust, and personalization use cases.

Aeva, Inc

Mountain View, CA

Software Engineering Intern — Perception

Aug 2022 – Dec 2022 | Jul 2023 – Dec 2023

→ Automated querying trims of data from a log recorded from an ARIES 2 sensor based on a scene description.

→ Implemented self-supervised training to use native ARIES 2 data to classify objects for autonomous driving.

→ Worked with translating synthetic data of specific scenes to a trainable MMLAB format to further improve model performance on pedestrian data.

Amazon

Seattle, WA

Applied Scientist Intern

May 2022 – Aug 2022

→ Designed a neural-network-based federated learning framework that predicts buyer fraud across regions, abiding by local data protection regulations.

Carnegie Mellon University

Pittsburgh, PA

Research Assistant (Synergy Labs)

Jan 2020 – Mar 2021

→ Built AppService, an infrastructure to host and run IoT apps in discrete LXC containers with dynamic resource allocation.

→ Designed several REST APIs for external systems to deploy, manage and interact with third-party apps on AppService.

→ Interfaced AppService with Building Depot, a Smart-Building OS to interact with 500+ sensors across CMU campus.

→ Worked on a privacy wrapper that restricts third party API calls in an app, and enforces User Access Control.

→ Developed IoT apps that use Machine Learning to make real-time inference on sensor data with minimal overhead.

Solarillion Foundation

Chennai, India

Undergraduate Research & Teaching Assistant, Server Administrator

Apr 2018 – May 2020

→ Proposed a scalable deep learning solution for stress detection on edge devices, achieving 95.39% computational savings — a 10% absolute accuracy increase over the benchmark and an 18x reduction in inference times on a Raspberry Pi 3.

→ Led a team of 4 and worked with real-world theatre occupancy data to build a two-stage model that predicts the number of weeks a movie is expected to screen based on behavioral population analysis; solution used by a top multiplex in India.

SELECTED PUBLICATIONS

1. **REMI: A Novel Causal Schema Memory Architecture for Personalized Lifestyle Recommendation Agents**
 PDF  arXiv
Workshop paper at **KDD Workshop on Online and Adaptive Recommender Systems (OARS'25)**, Toronto, Canada (Aug 6, 2025)
2. **Bayesian Active Learning for Wearable and Mobile Health**  Slides/Poster
NeurIPS Europe Meetup on Bayesian Deep Learning (BDL'20), December 2020 — Poster/Talk
3. **Bayesian Active Learning for Wearable Stress and Affect Detection**
Poster at **NeurIPS Workshop on Machine Learning for Mobile Health (ML4MH) 2020**
4. **A Two-Stage Machine Learning Approach to Forecast the Lifetime of Movies in a Multiplex**  Code
FICC 2020, San Francisco, USA
5. **Scalable Deep Learning for Stress and Affect Detection on Resource-Constrained Devices**  Slides
 Code
ICMLA 2019, Florida, USA
6. **Diagnosing Abnormality of Foetus Using Machine Learning Algorithms**
ICIOT 2019 — International Conference on Internet of Things, SRM Institute of Science and Technology, Chennai, India (Mar 11–15, 2019)

SELECTED PROJECTS

ETHViz: Ethereum Network Topology Visualization & Private-Net GUI  Code 2023
→ Discovered peers from a custom `geth` light node via `Web3's admin.peers()`, geolocated IPs, and rendered interactive maps/graphs (React, Chart.js, D3-geo, React-simple-maps, React-force-graph-3d) backed by a Flask service on AWS EC2; also built a GUI to connect peers/send transactions on private networks.

BERT Siamese Networks for Similar-Intent Question Identification  Code  Project Blog Oct 2021
→ Fine-tuned transformer backbones (BERT, GPT-2, XLNet) on Quora Question Pairs (404k pairs) with transitive augmentation (+25% data); final BERT model reached ~86% test accuracy. Built an unsupervised pipeline (BERT embeddings → PCA to 15 comps, K-means ~100 clusters) to narrow candidates, yielding ~100× faster inference for duplicate detection at scale.

WaDeNet: Wavelet Decomposition based CNN for Speech Processing  arXiv Nov 2020
→ Proposed an end-to-end speech model that embeds discrete wavelet decomposition within the CNN, enabling the network to learn multi-resolution spectral features without separate feature pipelines; reported an average +6.36% accuracy gain over prior baselines on mobile-health speech tasks with reduced model size/compute.

Transfer Learning for International Crisis Response  Leaderboard  Code Jan 2020
→ Used RoBERTa to transfer knowledge across organizations for crisis-tweet classification; focused on boosting performance for data-sparse orgs; **placed 5th** on the AMLD 2020 AICrowd challenge leaderboard.

Pollen Grain Classification  Google Scholar  Code May 2020 – Jun 2020
→ Implemented a two-stage pipeline using U-Net (segmentation) and VGG-16 (classification) to classify microscope pollen-grain images; entry to the ICPR 2020 Pollen Grain Classification Challenge.

TALKS & PRESENTATIONS

→ **KDD 2025 — OARS Workshop (Toronto, Aug 6, 2025)**: Presented “REMI: A Novel Causal Schema Memory Architecture for Personalized Lifestyle Recommendation Agents.”  PDF

→ **NeurIPS 2020 — ML4MH Workshop**: Poster — “Bayesian Active Learning for Wearable Stress and Affect Detection.”  arXiv

→ **NeurIPS Europe 2020 — Bayesian Deep Learning Meetup**: Talk/Poster — “Bayesian Active Learning for Wearable and Mobile Health.”  Event  Slides

→ **FICC 2020 (San Francisco)**: Presented paper — “A Two-Stage Machine Learning Approach to Forecast the Lifetime of Movies in a Multiplex.”

→ **ICMLA 2019 (Florida)**: Presented paper — “Scalable Deep Learning for Stress and Affect Detection on Resource-Constrained Devices.”

→ **ICIOT 2019 (Chennai)**: Paper presentation — “Diagnosing Abnormality of Foetus Using Machine Learning Algorithms.”  Conference

PEER REVIEW & PROGRAM COMMITTEES

- ICML 2026 — Foundation Models for Structured Data workshop (FMSD @ ICML 2026) (Reviewer). [Workshop](#)
- ICCV 2025 — The Second Workshop on Multimodal Representation and Retrieval (MRR 2025) (Reviewer). [Workshop](#)
- KDD 2025 — Workshop on Prompt Optimization (Reviewer). [Workshop](#)
- KDD 2025 — The 2nd Workshop on Large Language Models for E-Commerce (LLM4ECommerce) (Reviewer). [Workshop](#)
- KDD 2025 — Structured Knowledge for Large Language Models (SKnow-LLM) (Reviewer). [Workshop](#)
- KDD 2025 — SciSoc LLM Workshop: Large Language Models for Scientific and Societal Advances (Reviewer). [Workshop](#)
- AMLC 2025 — Main Conference (Reviewer). [About AMLC](#)
- AMLC 2025 — 5th Workshop on Machine Learning on Graphs (MLog) (Reviewer). *Internal workshop; no public link available.*
- ICMLA 2020 — Special Session on Deep Learning (Reviewer). [Special Sessions](#)
- AAAI-26 — Workshop on Health Intelligence (W3PHIAI-26) (Program Committee Member).

SKILLS

Languages: Python, C/C++, SQL, Bash

Core ML: transformers (tabular & sequence), gradient-boosted trees (XGBoost/LightGBM), reinforcement learning (GRPO/PPO), anomaly detection, self-supervised representation learning, Bayesian active learning, calibration & thresholding

Graph & Network ML: entity/interaction graphs, knowledge graphs, temporal graphs (Temporal GNNs), multi-hop relational reasoning, link prediction, community detection; tools: NebulaGraph/NGQL, PyTorch Geometric, DGL, NetworkX

LLMs & GenAI: agentic AI systems, agentic RAG / retrieval-augmented generation, multi-agent orchestration, LLM pipeline design, parameter-efficient fine-tuning (QLoRA), prompt design & evaluation, Amazon Bedrock (agents, knowledge bases, guardrails)

Production ML: large-scale feature pipelines (logs / graphs / clickstream), offline & online evaluation, A/B testing, drift monitoring, low-latency model serving under strict SLAs

Tools & Cloud: PyTorch, Hugging Face Transformers, scikit-learn, Pandas/NumPy, Docker; AWS (Bedrock, SageMaker, EC2, S3, Lambda)

ACHIEVEMENTS AND EXTRA-CURRICULARS

- Recipient of Amazon's inaugural (first-edition) **VP-level GenAI “Just Did It” Trailblazer Award** (May 2025) for high-impact generative-AI innovation in fraud prevention.
- Promoted from Applied Scientist I to Applied Scientist II (Oct 2025).
- Submission ranked in top 3% in the Spotify Sequential Skip Prediction challenge conducted by Spotify and AICrowd.
- Best project award at SRM ICIOT 2019.
- Led the blogging team at SRM Alumni association.